

Newton-Raphson Method

A fast iterative method for solving nonlinear equations, $f(x) = 0$

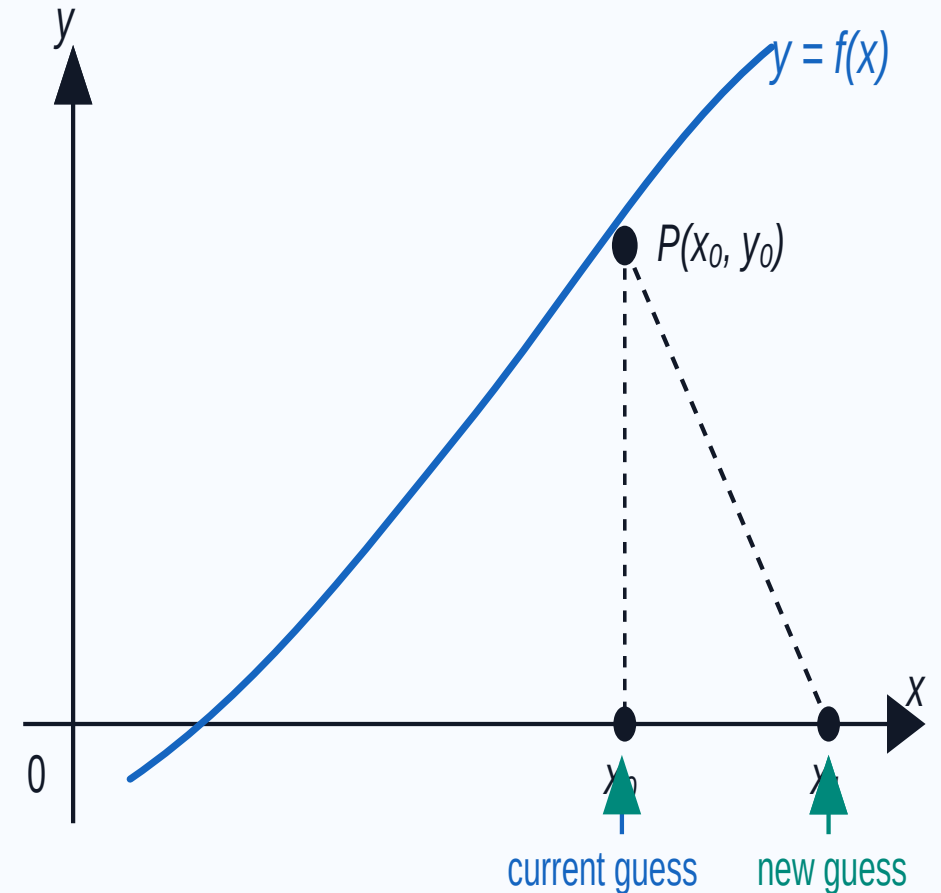
Main message

Use the tangent line at the current guess to jump closer to the root.

Formula to remember

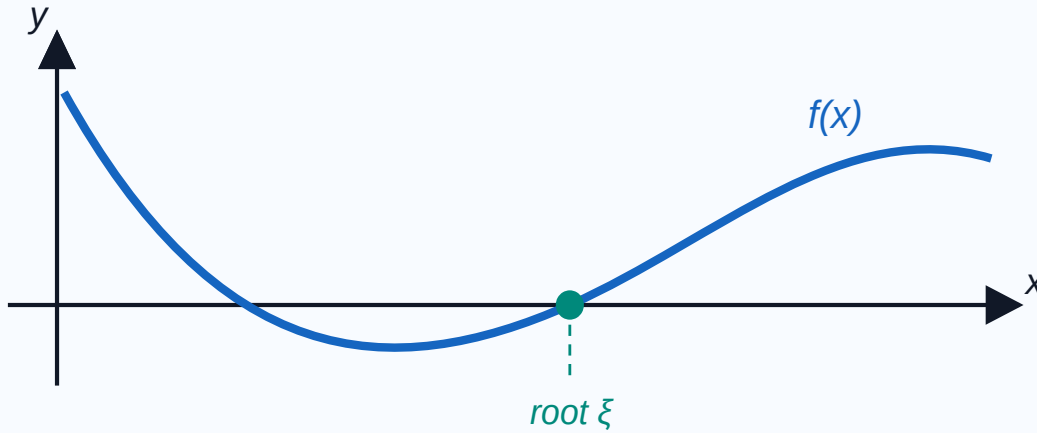
$$x_{n+1} = x_n - f(x_n) / f'(x_n)$$

Repeat until the approximation changes only slightly.



1. What problem are we solving?

We want the x -value where the function becomes zero.



Root of an equation

A root is a value ξ such that:

$$f(\xi) = 0$$

Graphically, it is the point where the curve cuts the x-axis.

Why use Newton-Raphson?

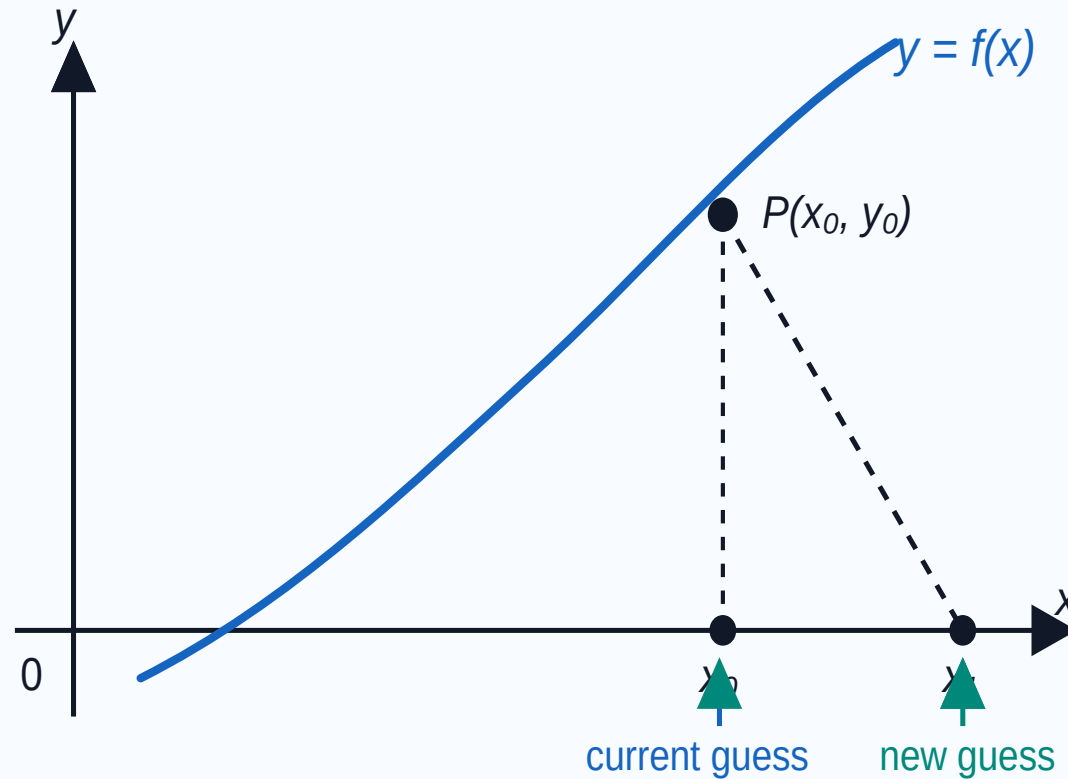
- Exact algebraic solutions are not always possible.
- The method gives a fast numerical approximation.
- It is widely used in engineering, science, and computing.

Simple target

Start with a guess, then improve it step by step until $f(x)$ is almost zero.

2. Geometrical intuition

Each iteration uses a tangent line to create the next guess.



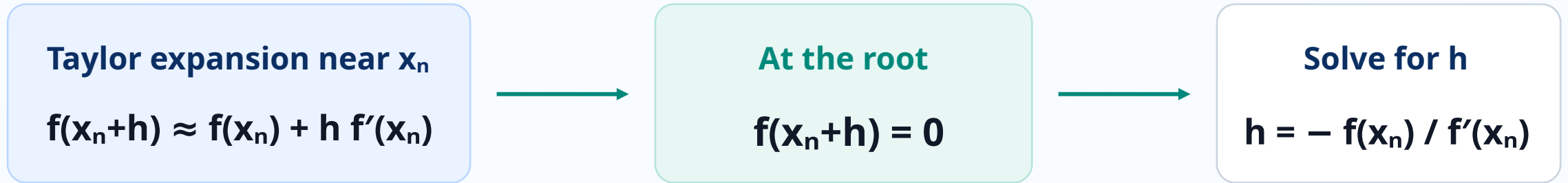
How the tangent helps

- 1 Start at the current point $P(x_n, f(x_n))$.
- 2 Draw the tangent to the curve at that point.
- 3 Find where this tangent cuts the x-axis.
- 4 Use that x-intercept as the next guess x_{n+1} .

Better approximation: $x_n \rightarrow x_{n+1}$

3. Derivation from Taylor series

The formula comes from a first-order approximation near the current guess.



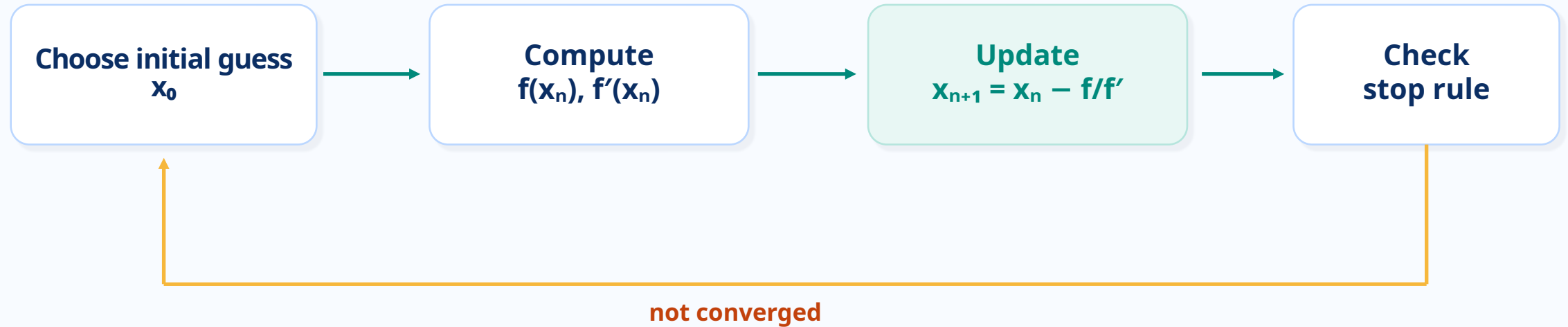
Final Newton-Raphson update

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \mathbf{h} = \mathbf{x}_n - \mathbf{f}(\mathbf{x}_n) / \mathbf{f}'(\mathbf{x}_n)$$

Key assumption: h is small, so higher-order terms are ignored.

4. Algorithm steps

Apply the same update again and again until convergence.



Common stop rules

$$|x_{n+1} - x_n| < \varepsilon \quad \text{or} \quad |f(x_{n+1})| < \varepsilon$$

Important condition

Do not divide by zero, $f'(x_n)$ must not be 0.

5. Worked example: Find $\sqrt{2}$

Rewrite the problem as $f(x)=0$, then iterate.

Set up the equation

$\sqrt{2}$ is the positive root of

$$f(x) = x^2 - 2, \quad f'(x) = 2x$$

Iteration table

n	x_n	$f(x_n)$
0	1.5000	0.2500
1	1.4167	0.0069
2	1.4142	≈ 0.0000
3	1.4142	≈ 0.0000

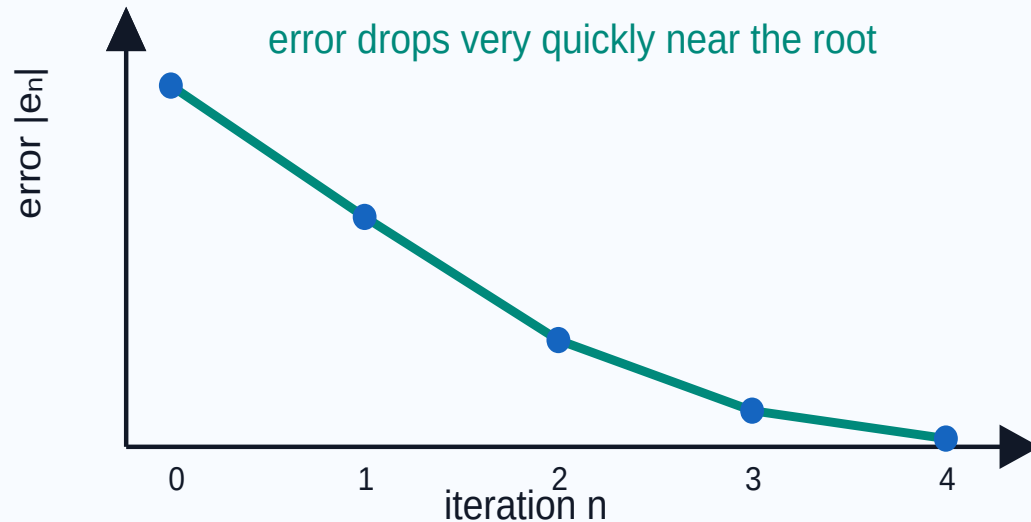
Result after only a few iterations

$$\sqrt{2} \approx 1.4142$$

Starting guess used: $x_0 = 1.5$

6. Convergence and limitations

Newton-Raphson is very fast near a simple root, but it is not guaranteed from every guess.



Quadratic convergence

Near a simple root, the next error is roughly proportional to the square of the current error:

$$e_{n+1} \approx C e_n^2$$

This is why the number of correct digits can grow quickly.

Main cautions

- The derivative $f'(x_n)$ must not be zero.
- A poor starting guess can move away from the desired root.
- Multiple roots often reduce the convergence speed.

7. Summary

The essential points to remember.

Newton-Raphson in one sentence

It improves an initial guess by replacing the curve with its tangent line and using the tangent x-intercept as the next guess.

Formula

$$x_{n+1} = x_n - f(x_n)/f'(x_n)$$

1 Start

Choose an initial guess close to the root.

2 Iterate

Apply the Newton-Raphson update repeatedly.

3 Stop

Finish when the change or $f(x)$ is very small.

Best used when the function is smooth and the starting guess is reasonably good.